EMMANUEL SEUN SHITTU

Backend Engineer

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Backend Engineer with years of experience building secure, high-throughput systems for payments, fintech and multi-tenant SaaS platforms using TypeScript, Node.js, Express and NestJS. Hands-on with payment rails (Paystack, Monnify, crypto gateways), signed and idempotent webhook processing, RBAC-secured APIs and Redis-backed asynchronous workflows. Also delivers React, Next.js and React Native (Expo) front ends. Linux Foundation KCNA certified, with AWS Cloud Practitioner training in progress.

TECHNICAL SKILLS

Backend & Architecture	TypeScript, Node.js, Express.js, NestJS, REST APIs, gRPC, Microservices, Event-Driven Systems, WebSockets / Socket.IO
Payments & Integrations	Paystack, Monnify, Crypto Gateways, Webhook Processing (HMAC signature verification), Idempotent API Design, Reconciliation
Databases & Messaging	PostgreSQL, MySQL, MongoDB, Redis (caching and queues), BullMQ, RabbitMQ, Query Optimization
Cloud & DevOps	Docker, Kubernetes (KCNA certified), GitHub Actions (CI/CD), AWS (EC2, S3), Render, Linux
Security	JWT, Refresh-Token Rotation, RBAC, OAuth2, Rate Limiting, Secure API Design
Frontend & Mobile	React.js, Next.js, React Native (Expo), TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS
Testing & Tools	Jest, Supertest, Swagger (OpenAPI), Postman, Git/GitHub, Agile/Scrum

PROFESSIONAL EXPERIENCE

Divnerix Innovations | Remote, Nigeria

Software Engineer **12/2025 – Present**

- Build cross-platform mobile applications with React Native and Expo, including fintech and geospatial features, integrated with the backend APIs.

Backend Engineer **01/2025 – 11/2025**

- Owned backend systems for live payment, donation and event platforms built on TypeScript, Node.js and Express.js.
- Designed and secured REST APIs handling real transactions across Paystack and crypto payment gateways, including webhook-based payment confirmation.
- Implemented JWT/RBAC authentication and authorization across multi-tenant, organization-based systems.
- Built event-driven asynchronous workflows on Redis queues for reliable background processing at scale.
- Improved API performance by up to 40% through caching, query optimization and efficient data modeling under high traffic.
- Set up CI/CD pipelines and managed deployments on AWS infrastructure.

Backend Lead (Volunteer) | ScholarGuide (EdTech Startup), Remote **02/2025 – 10/2025**

- Led backend architecture and API design for a multi-role EdTech platform (students, parents, teachers, admins) built with TypeScript.
- Built a gamification engine (points, badges, leaderboards) and multi-role JWT/RBAC authentication.
- Designed data models across PostgreSQL, MongoDB and MySQL, with Redis caching, indexing and query tuning.
- Built a search and recommendation system that improved query performance by up to 75%.
- Implemented rate limiting and secure API practices; documented APIs with Swagger for team collaboration.

Full Stack Developer | In-Box Elite Nig. Ltd (Merchandising), Benin City **02/2022 – 12/2024**

- Built backend services for product, order and inventory management using TypeScript, Node.js, Express and MySQL.
- Designed RESTful APIs for business operations and automated internal workflows, improving efficiency by 40%.
- Improved API performance by 30% and delivered full-stack integration for operational business systems.
- Introduced real-time low-stock alerts, reducing stockouts and manual stock checks.
- Designed an order-tracking dashboard giving the sales team live visibility into order status, cutting customer inquiry response time.
- Implemented a caching layer for frequently accessed product data, reducing average API response times.

Full Stack Developer (Intern) | Edo Job, Benin City **05/2022 – 12/2022**

- Built and shipped 7+ full-stack applications end to end using React, TypeScript, Node.js, Express and MongoDB, from requirements through deployment.
- Developed and maintained REST APIs for live client-facing features; worked in Agile sprints with planning, stand-ups and code reviews.
- Deployed applications to cloud hosting platforms and managed production environment configuration.

Full Stack Bootcamp, Cohort 5 (Trainee) | Edo Innovate, Benin City **03/2021 – 12/2021**

- Completed intensive project-based training in React, TypeScript, Node.js, Express and MongoDB, with Git/GitHub workflows, pair programming and code review practice.

SELECTED PROJECTS

Nest E-commerce API

NestJS, TypeScript, PostgreSQL, TypeORM, Paystack, Jest, Docker | GitHub: nest-ecommerce-api

- Race-safe checkout using conditional stock updates inside a single transaction, with idempotency keys so retries never double-reserve stock.
- Paystack payments with HMAC-SHA512 signature-verified, replay-safe webhooks, amount validation and an order state machine (PENDING_PAYMENT to DELIVERED).
- JWT with rotating refresh tokens, role-based access, PostgreSQL full-text search and 24 end-to-end tests against a real database.

Monnify Payroll API

TypeScript, Node.js, Express, PostgreSQL, Redis, BullMQ, Docker | GitHub: monnify-payroll-api

- Bulk salary disbursement through Monnify from a BullMQ worker; items are claimed atomically so no employee is ever paid twice.
- Signed, deduplicated webhooks and automatic reconciliation of timed-out or lost payments; failed items retried under new payment references.
- JWT authentication with three roles (Admin, Payroll Manager, Viewer), masked account numbers and an audit log of sensitive actions.

Fintech Microservices Platform

NestJS, gRPC, PostgreSQL, RabbitMQ, Redis, BullMQ, Docker, Nx | GitHub: Fintech-Microservices-Platform

- Nx monorepo of five NestJS services (Auth, Wallet, Transaction, Integration, Notification) communicating over gRPC and RabbitMQ.
- Forex order orchestration with BullMQ retries (up to 3 attempts) for temporary failures; Redis-cached exchange rates refreshed by a daily cron job.

Payment Parser & Executor API

Node.js, JavaScript | GitHub: payment-parser-executor-API

- API that parses and executes structured payment instructions (e.g. debit and credit between accounts) with schema validation up front.
- Layered architecture (endpoint, middleware, service, repository) with centralized error codes, structured logging and path aliases.

ADDITIONAL PROJECTS

Support Ticket System API

Node.js, Express, MySQL, JWT | GitHub: Backend-Support-Ticket-System | Live API

- Public ticket submission endpoint plus JWT-protected admin endpoints to list, filter and update ticket status; deployed on Render.

Amortisation Engine

Node.js, Express, MongoDB, Docker, Nginx | GitHub: Amortisation-engine

- Loan amortisation with variable interest rates and leap-year daily interest accrual; produces full repayment schedules.
- JWT authentication, bcrypt password hashing, role-based access, rate limiting and idempotency middleware.

Newsletter & Support API

Node.js, Express, MongoDB, Joi, Nodemailer | GitHub: Newsletter-Subscription-API

- Newsletter subscribe/unsubscribe and support inquiry endpoints with duplicate prevention, Joi validation and automated confirmation emails; MVC structure with Swagger docs.

Social Network Blog Backend

Node.js, Express, MongoDB, Socket.IO | GitHub: social-network-blog-backend

- REST API for authentication and social feed management, with real-time communication over Socket.IO, image serving, Helmet security headers and request logging.

OPEN SOURCE & CONTRIBUTION

Content Automation & Analytics Platform (Groovinx / SkyeStudio)
TypeScript, Node.js, FFMPEG

- Built an automated multi-format video transcoding pipeline with FFMPEG, cutting manual processing time significantly.
- Designed APIs and a rule-based scheduling engine for recurring and conditional content publishing.
- Contributed reusable automation modules to the shared codebase for use across other content projects.

EDUCATION

B.Sc. Banking and Finance | University of Benin, Benin City **2018**
National Diploma, Business Administration | Abraham Adesanya Polytechnic, Ijebu Igbo **2007 – 2010**

CERTIFICATIONS

- Kubernetes and Cloud Native Associate (KCNA), Linux Foundation
- Andela–CNCF Kubernetes African Developer Training Program, 2026
- AWS Certified Cloud Practitioner, in progress (AWS Training Program)
- Backend Development (Node.js), CareerEx
- Python and Statistics for Financial Analysis
- Foundations: Data, Data, Everywhere, Google